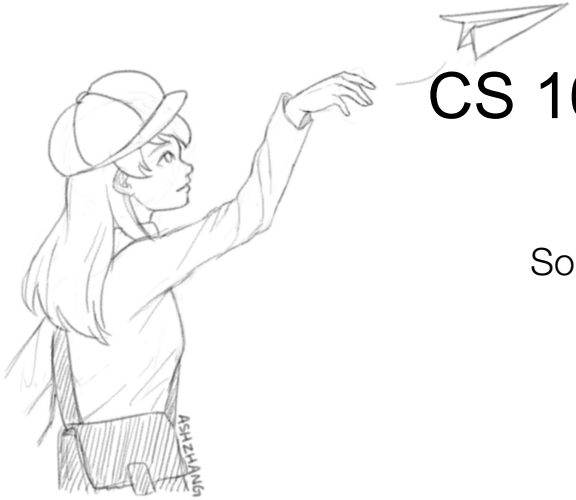
Note to self:
Set up terminal

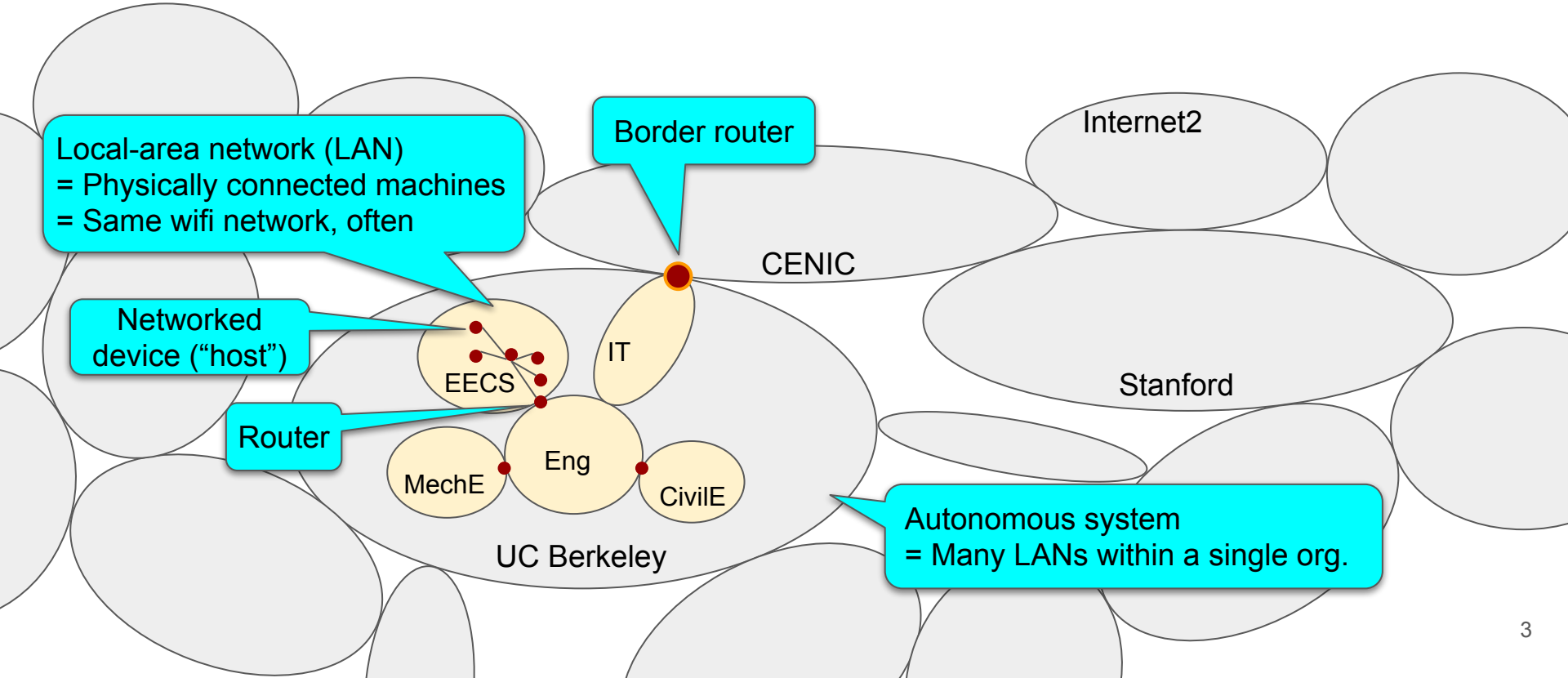
Introduction to Networking II

CS 161 Spring 2026 - Lecture 24



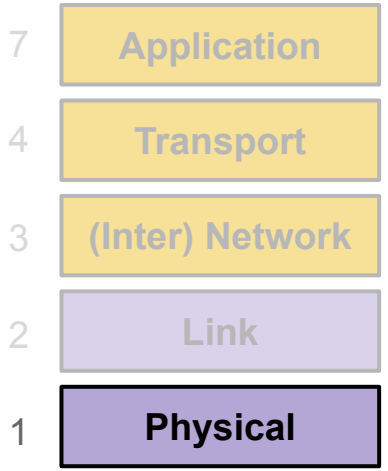
Some slides (as marked) are courtesy of UCB graduate and security expert Grant Ho.

Cartoon version of the Internet



Layer 1: Physical Layer

- **Provides**: Sending bits from one device to another
  - Encodes bits to send them over a physical link
    - Patterns of voltage levels
    - Photon intensities
    - RF modulation
- Examples
  - Wi-Fi radios (IEEE 802.11)
  - Ethernet voltages (IEEE 802.3)

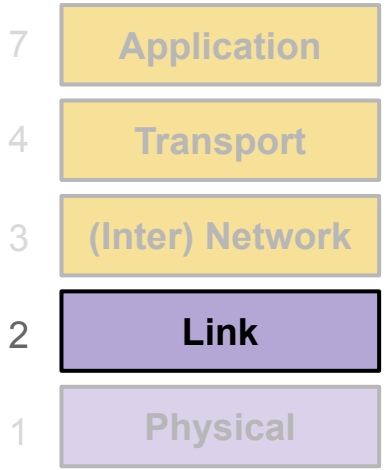


Layer 2: Link Layer

- **What it's for**: Sending bits within a LAN
  - **Makes use of**: Physical connection between LAN hosts
  - A chunk of bits at layer 2 is called a "frame"

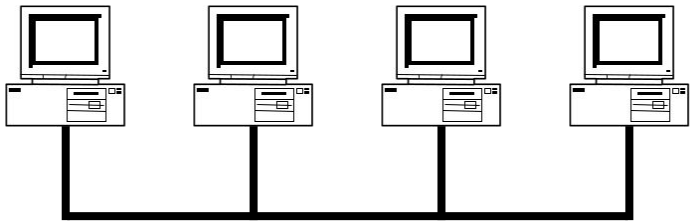
- Most common example: Ethernet
  - Standard frame has size ≤ 1500 bytes
  - If you want to send a long message, you send many frames

- Bizarre feature of Ethernet: "Best-effort" delivery
  - If you send a frame over Ethernet, maybe it gets there
  - Maybe not!
  - Seems bad, but this simplifies the design tremendously
  - Design is (IMO) absolutely elegant



Ethernet and MAC Addresses

- **MAC (Ethernet) address**: A 6-byte address that uniquely identifies an Ethernet device
  - 48 bits or 6 hex bytes: **13:37:ca:fe:f0:0d**
  - The first 3 bytes identify device manufacturer
  - The last 3 bytes are device-specific

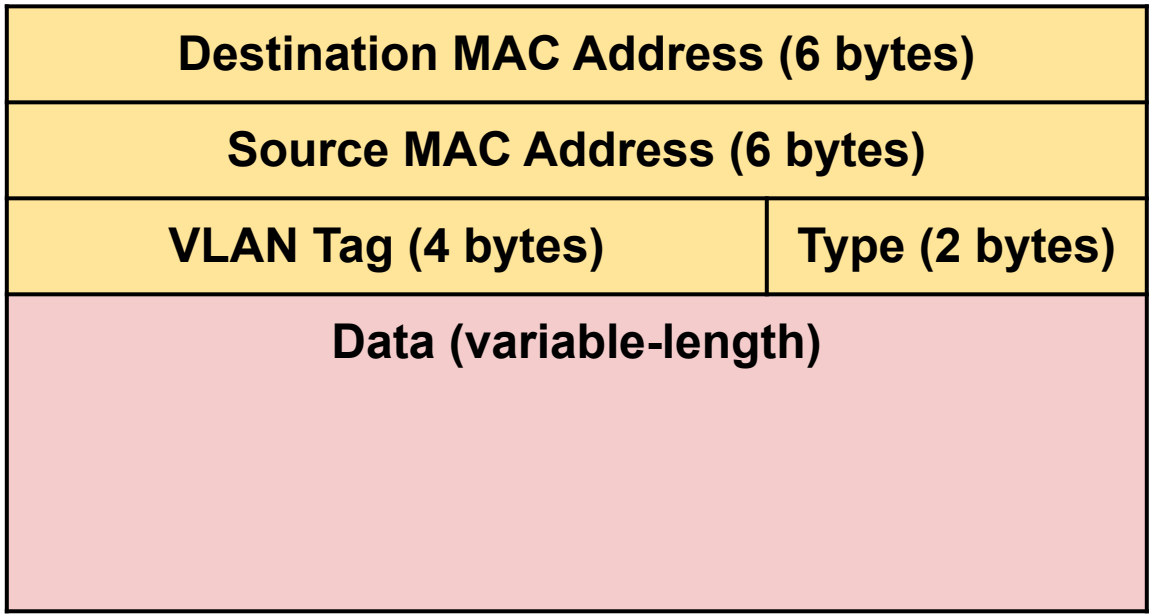


- A host can have many MAC addresses
  - E.g., for wired network card, WiFi interface

- MAC address is FIXED at manufacture time
  - But if you are sneaky, you can (usually) change it
  - Requires root on the machine

  ⇒ Don't rely on MAC for authentication plz!!!

Ethernet frame format



Slide courtesy of Grant Ho (UChicago) + my edits

Questions I got after lecture

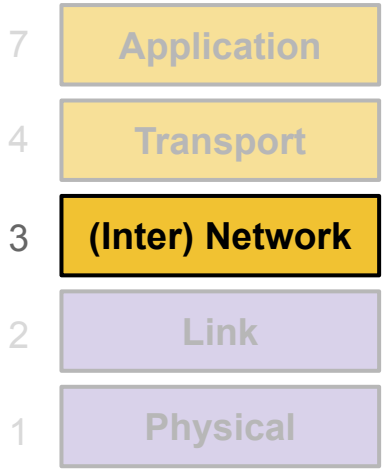
- When connecting to a web server in the same LAN do we still use IP?
  Or do we just use Ethernet?

- If every device in the world has a unique Ethernet address,
  why do we need IP addresses?

- When sending an IP packet across multiple LANs,
  does each LAN add and remove its own Ethernet frame info to the packet?

Layer 3: Network Layer

- **Used for:** send packets over Internet
  - **Relies upon**: Sending messages over LAN
  - Encodes messages into groups of bits called “packets”
    - Theoretically up to 2^{16}-1 bytes, actually 1500 bytes
    - Why?
  - **Bridges multiple LANs to provide global addressing**
- The only important one is the Internet Protocol (IP)
  - There used to be other ones, all essentially killed by IP



Internet Protocol (IP) addresses

- An address that identifies a device on the Internet

- IPv4 address is 32 bits
  - Typically written as 4 decimal octets, e.g. `35.163.72.93`
  - NOT fixed—assigned by network administrator when you connect to a network
  - Easy to spoof! ⇒ Do not use for serious authentication

- Someone in 1981: *"4 billion IPv4 addrs = more than we will ever need!"*

- IPv6 is 128 bits, typically written as 8 groups of 2 hex bytes: `2607:f140:8801::1:23`
  - If digits or groups are missing, fill with 0's, so `2607:f140:8801:0000:0000:0000:0001:0023`
  - In my experience, using IPv6 is a world of pain

UC Berkeley IP-address blocks

128.3.*.*

128.32.*.*

131.243.*.*

136.152.0.*-136.152.208.*

136.152.210.*-136.152.255.*

169.229.*.*

192.12.173.*

192.58.231.*

198.125.132.*-198.125.133.*

198.128.24.*-198.128.31.*

198.128.52.*

198.128.192.*-198.128.223.*

204.62.155.*

IP Packets

- Any IP router may fragment a packet into smaller packets
  - Different links might support different maximum packet sizes (Ethernet: 1500 bytes)
  - Up to the recipient to reassemble fragments into the original packet
- Each IP router forwards a given packet to the next hop
  - We will cover how a router knows how to forward—and attacks on it—in the future
- Packets are not guaranteed to take a given route
  - Two packets with the same source and destination may take different routes
- Want to send a big file? You can't with IP alone.
  - The API of the IP layer only supports short packets.

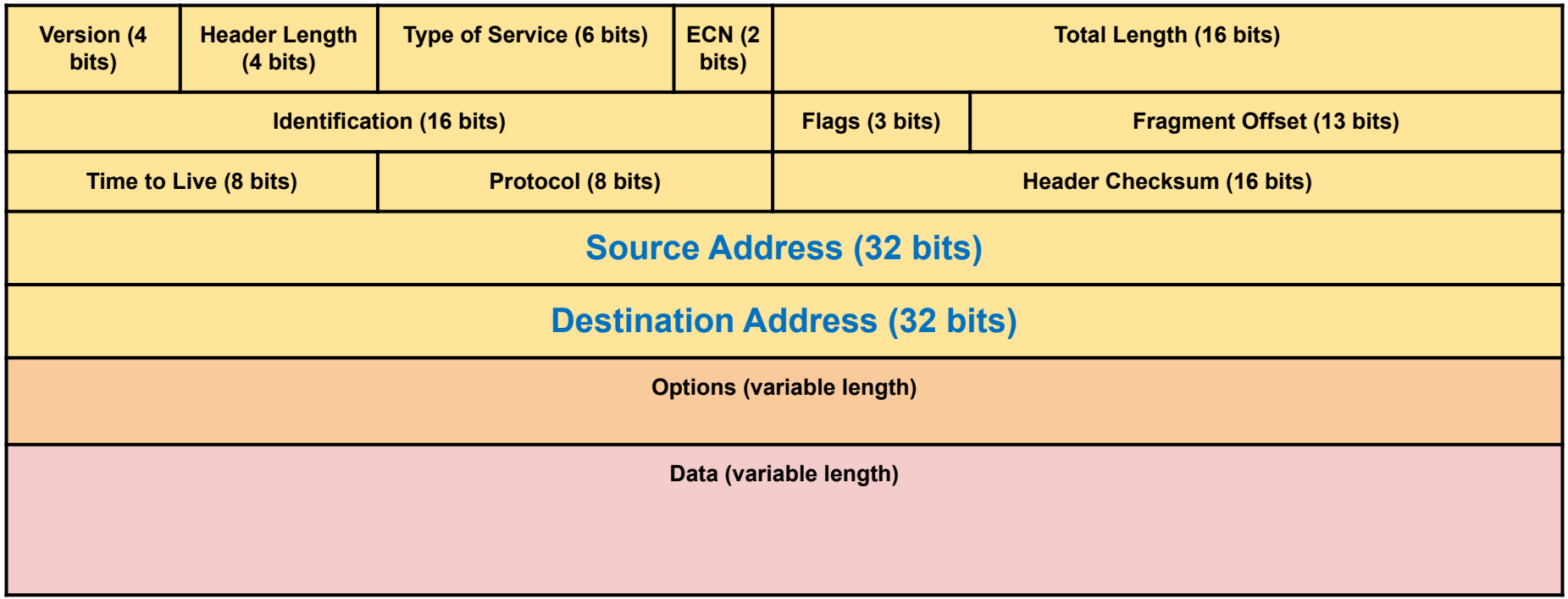
Properties of IP

- IP packets have a checksum to detect bitflips/corruption
  - Detects random corruption with probability $1 - 1/2^{32}$
  - Not a cryptographic MAC—there is no secret key
  - There are no guarantees if an attacker intentionally modifies packets

- IP is *unreliable*. Only provides a **best-effort** delivery service, which means:
  - Packets may be lost (“dropped”)
  - Packets may be corrupted
  - Packets may be delivered out of order

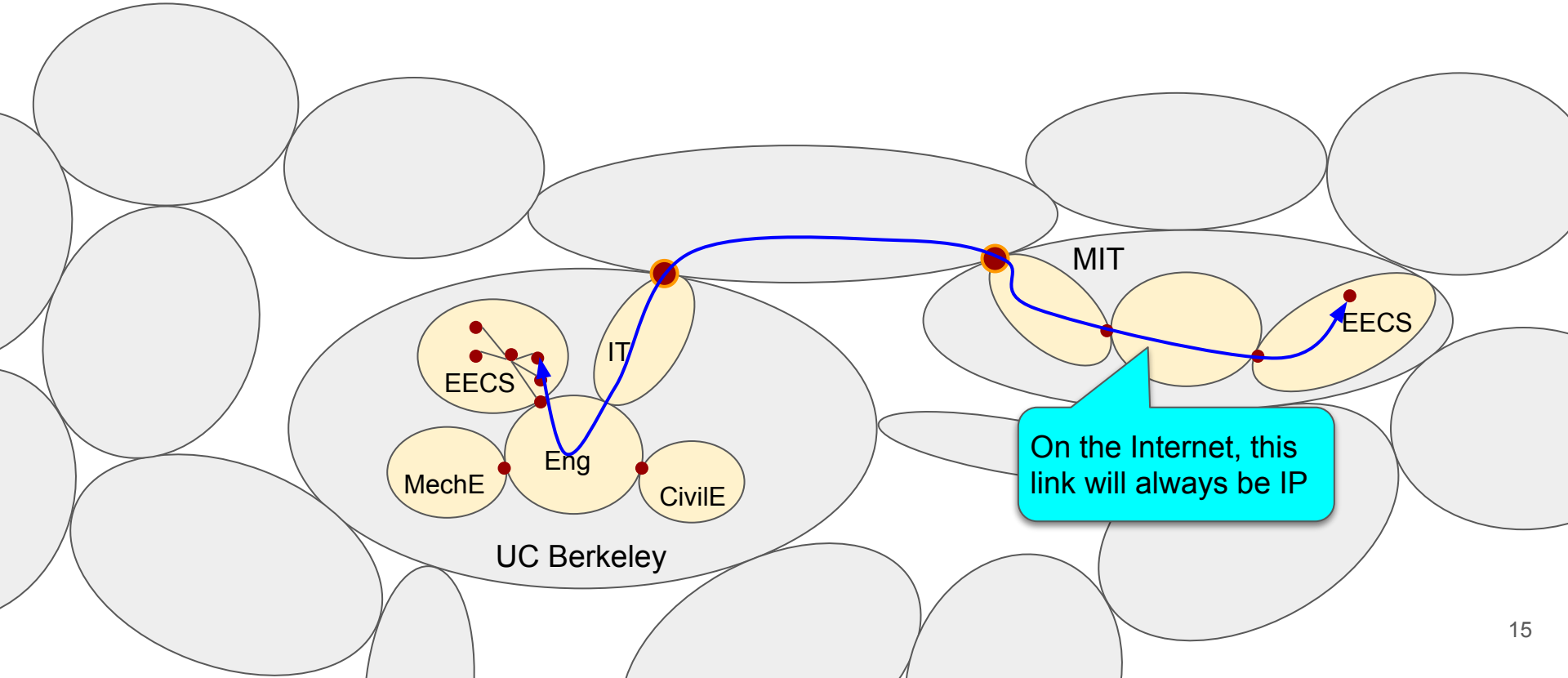
- It is up to higher level protocols to provide reliable transport

IPv4 packet format



Slide courtesy of Grant Ho (UChicago) + my edits

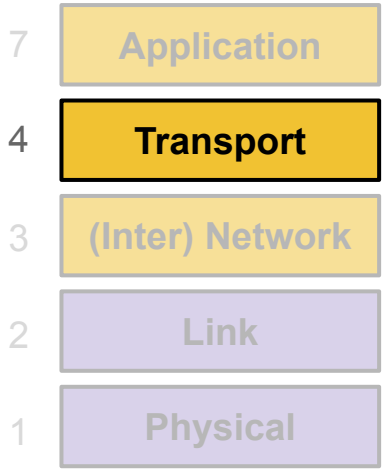
Back to our cartoon version of the Internet



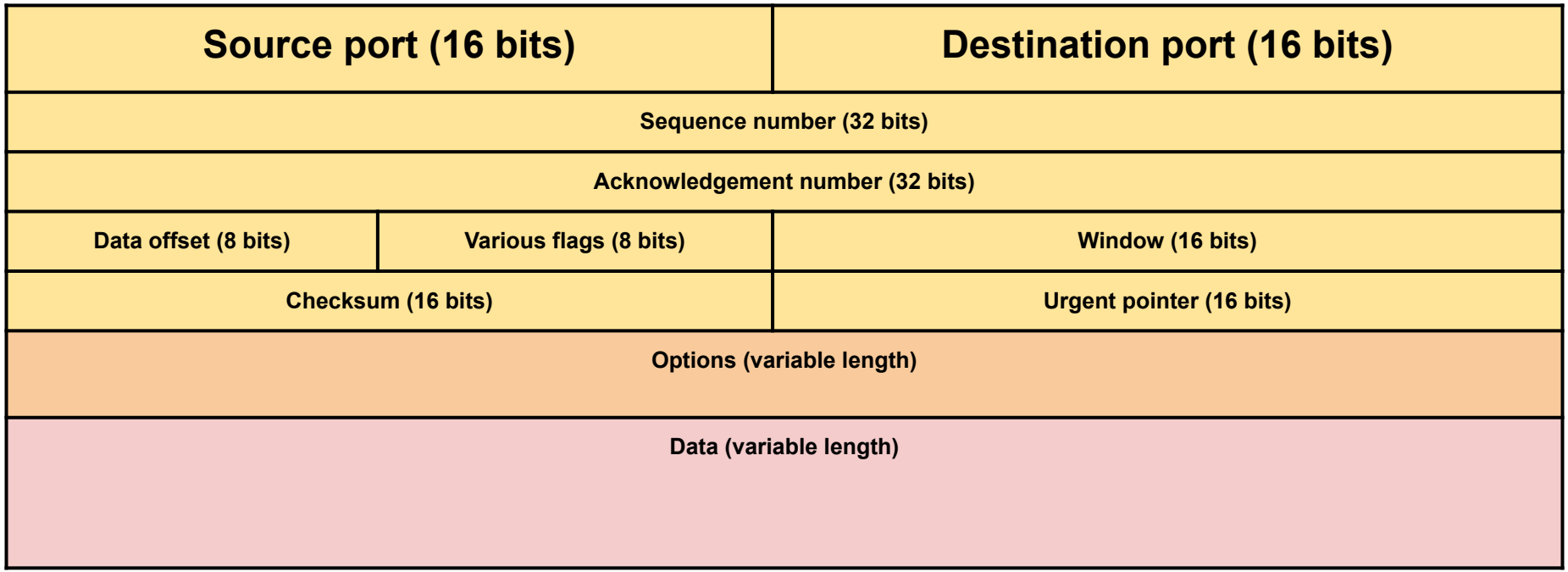
Show `mtr`

Layer 4: Transport Layer

- **Used for**: Transportation of arbitrary-length data from any point to any other point
  - **Relies upon**: Sending packets between devices
- Useful abstractions
  - **Reliability**: Transmit stream of bytes reliably, in order
  - **Ports**: Provide multiple “addresses” per real IP address
- Examples
  - **TCP**: Provides reliability and ports
  - **UDP**: Provides ports, but no reliability

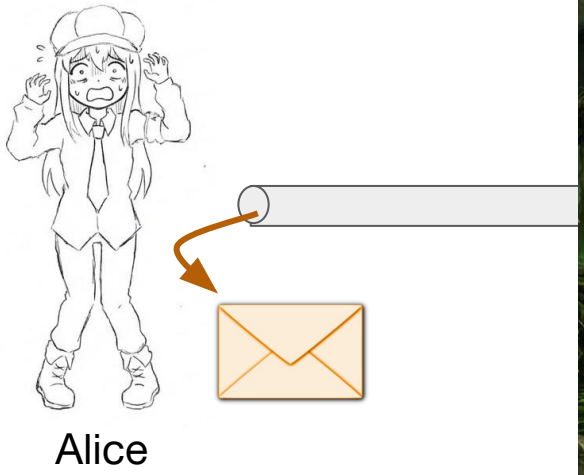


TCP segment format



Slide courtesy of Grant Ho (UChicago) + my edits

How you think about a TCP connection after CS161



Alice

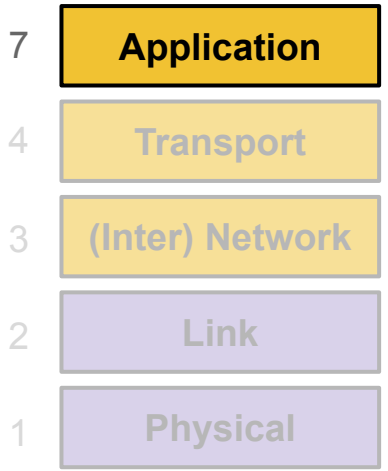


Show nc

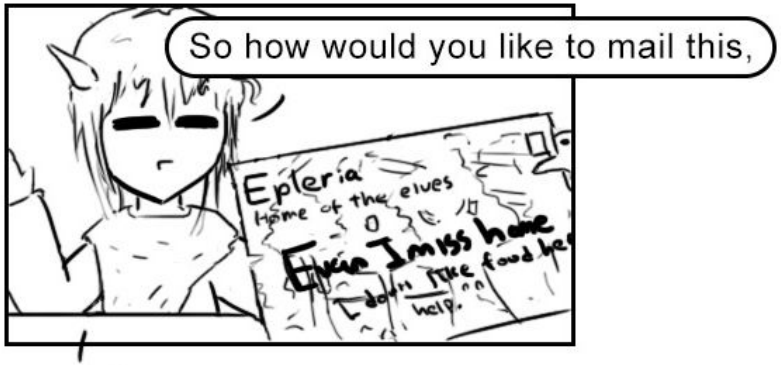
Layer 7: Application Layer

- **Used for**: End-user applications and services
  - **Relies upon**: Transportation of variable-length data from any point to any other point

- Most applications we use run on Layer 7 protocols:
  - Web browsing (HTTP)
  - Non-web email clients (POP, IMAP, SMTP)
  - Remote shell (SSH)
  - Messaging services (often HTTP)
  - Video calls (often HTTP)
  - Online video games (custom layer-7 protocols)



Headers


So how would you like to mail this,
Epteria
home of the elves
I miss home

Example: Sending Mail

Alice



I am hungry.

Bob



Example: Sending Mail

Alice





Send to: Bob

I am hungry.

Bob





Example: Sending Mail

Alice







Mail to: 123 Bob St

Send to: Bob

I am hungry.

Bob







Example: Sending Mail

Alice







Bob





Mail to: 123 Bob St

Send to: Bob

I am hungry.



Example: Sending Mail

Alice







Bob



Send to: Bob

I am hungry.





Example: Sending Mail

Alice







Bob

I am hungry.







Example: Sending Mail

Each layer communicates with each other, relying on abstractions below them!

Alice

Bob

Relies upon: Sending messages to people





Provides: Sending messages to people
Relies upon: Sending messages to addresses





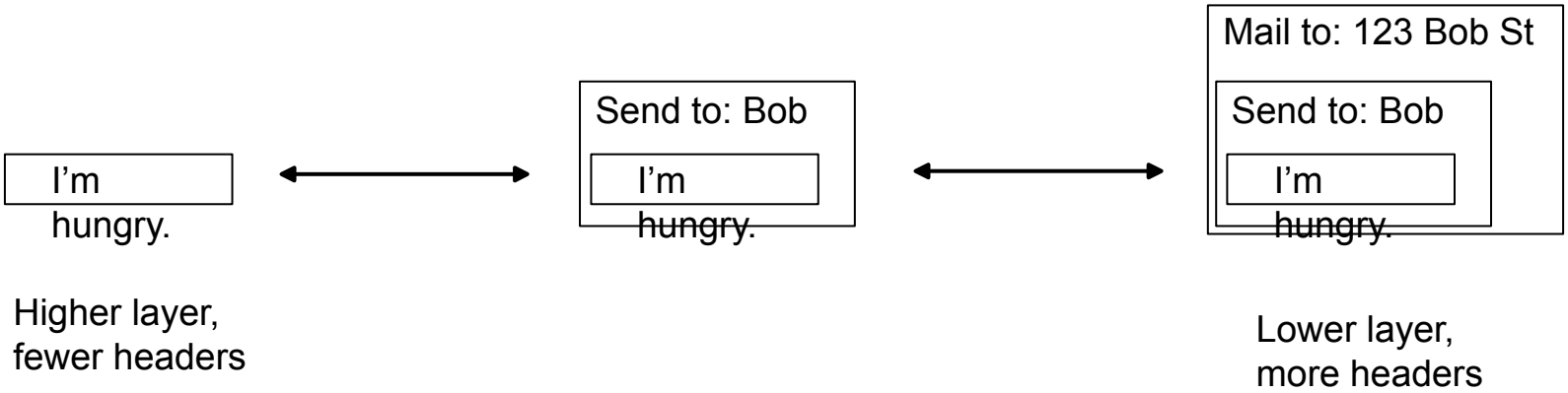
Provides: Sending messages to addresses



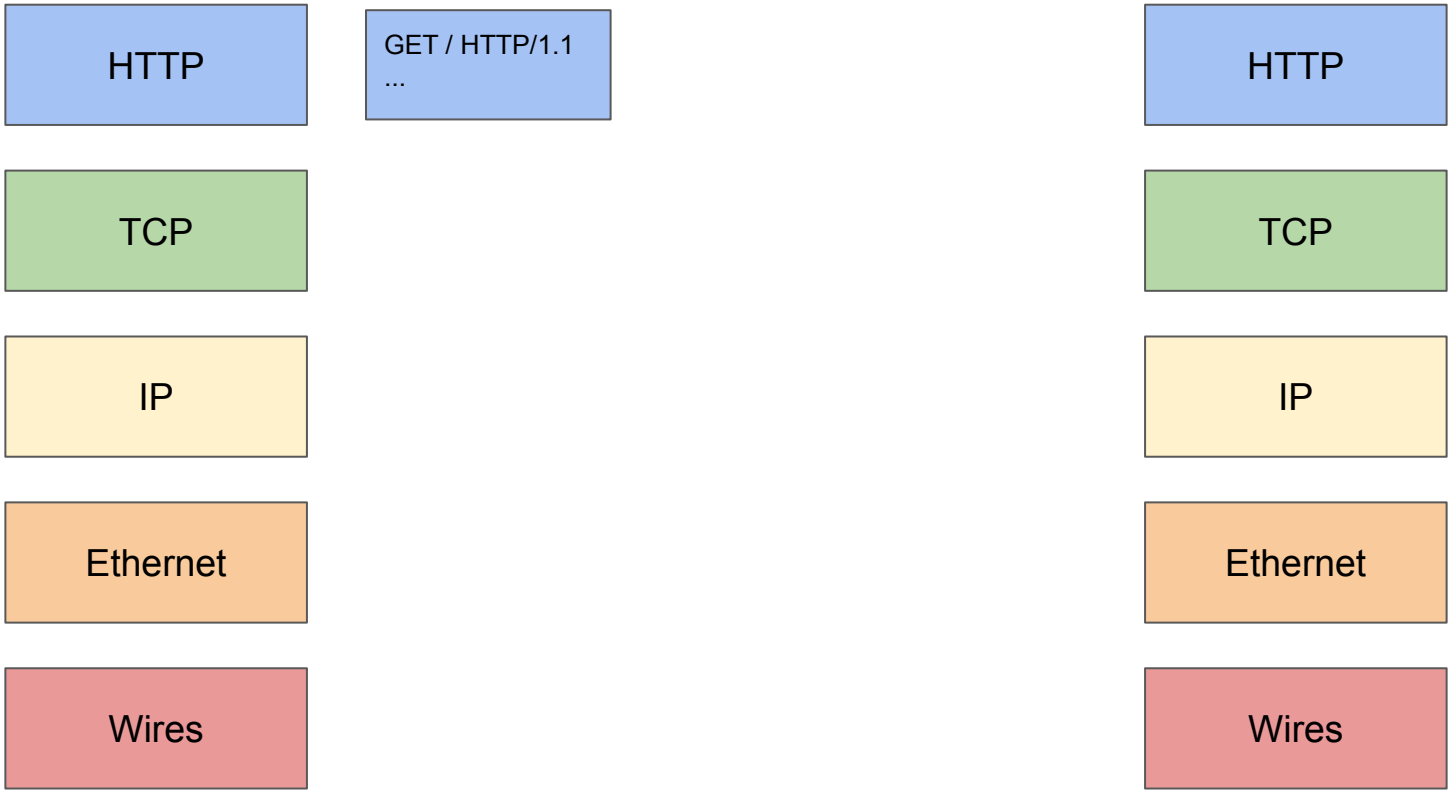


Layers of Abstraction and Headers

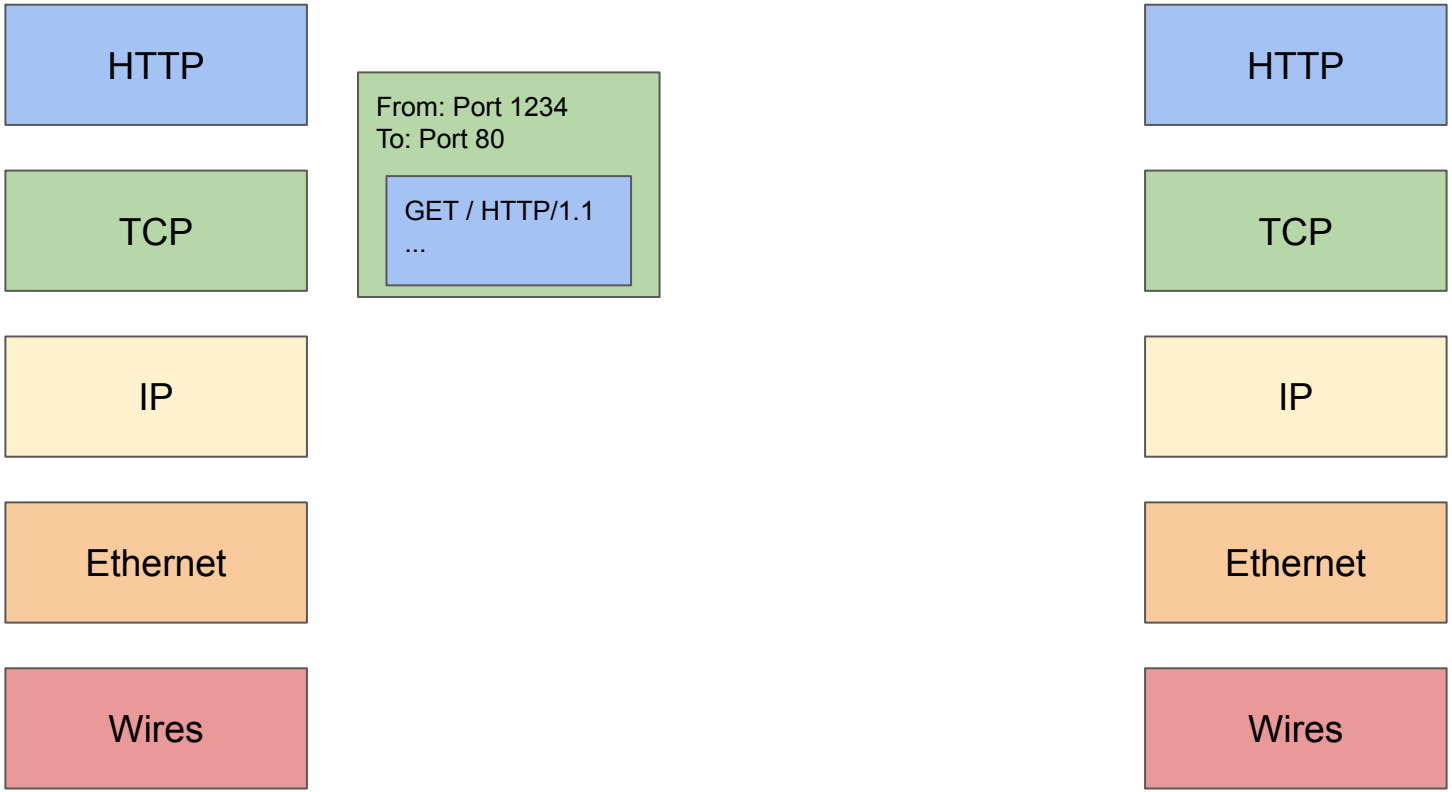
- Each protocol adds its own headers around the message
- Before delivering message to higher layer, protocol strips off header



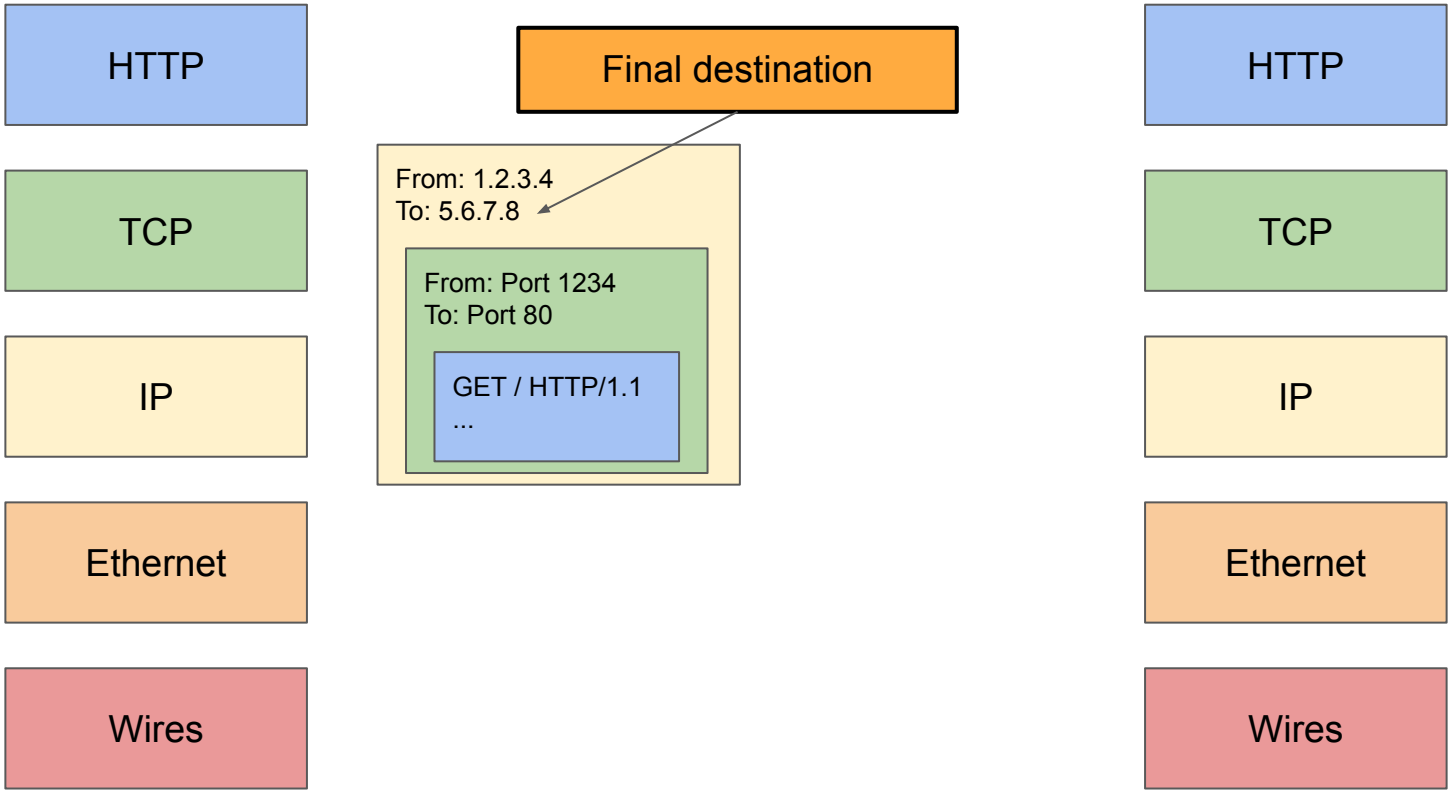
Example: HTTP Request



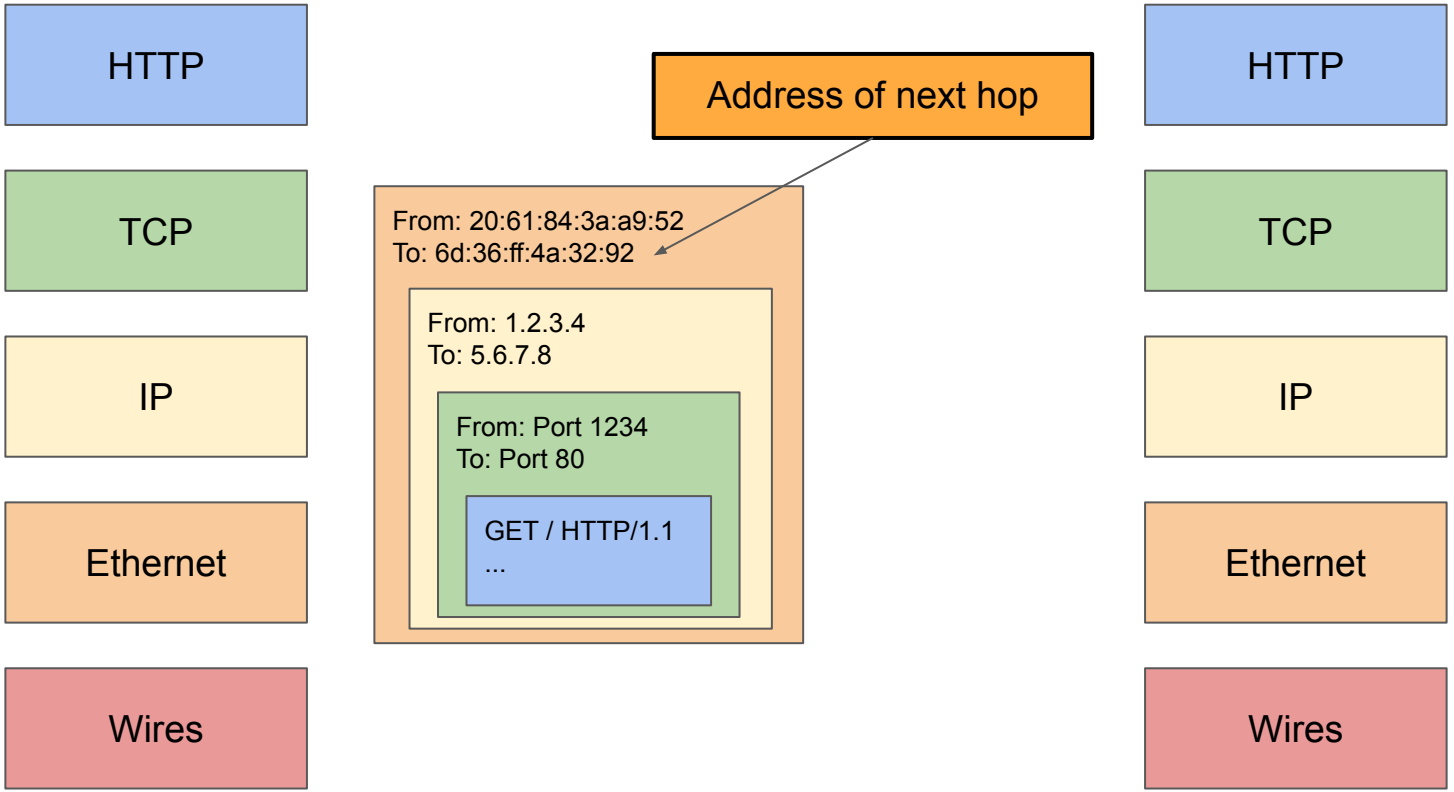
Example: HTTP Request



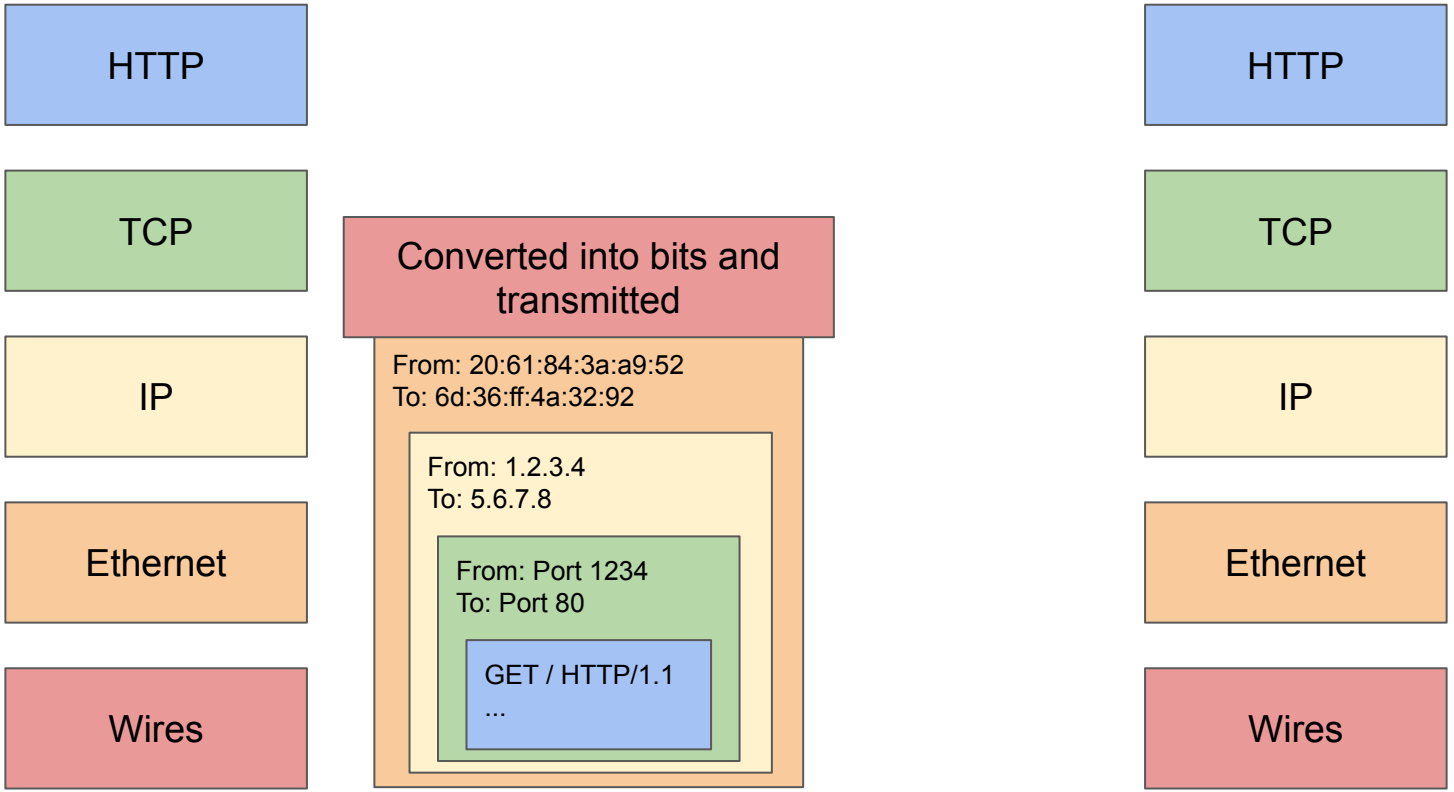
Example: HTTP Request



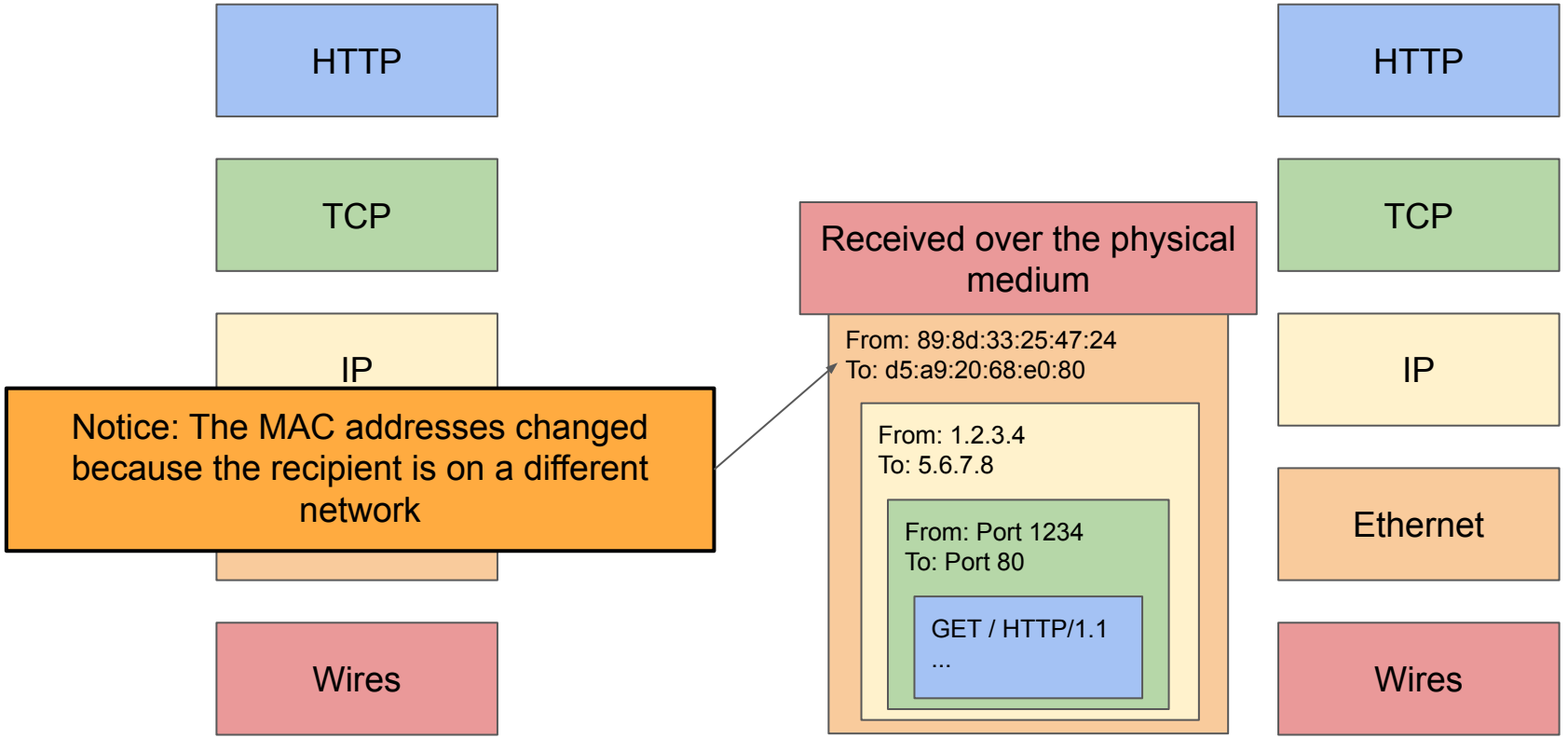
Example: HTTP Request



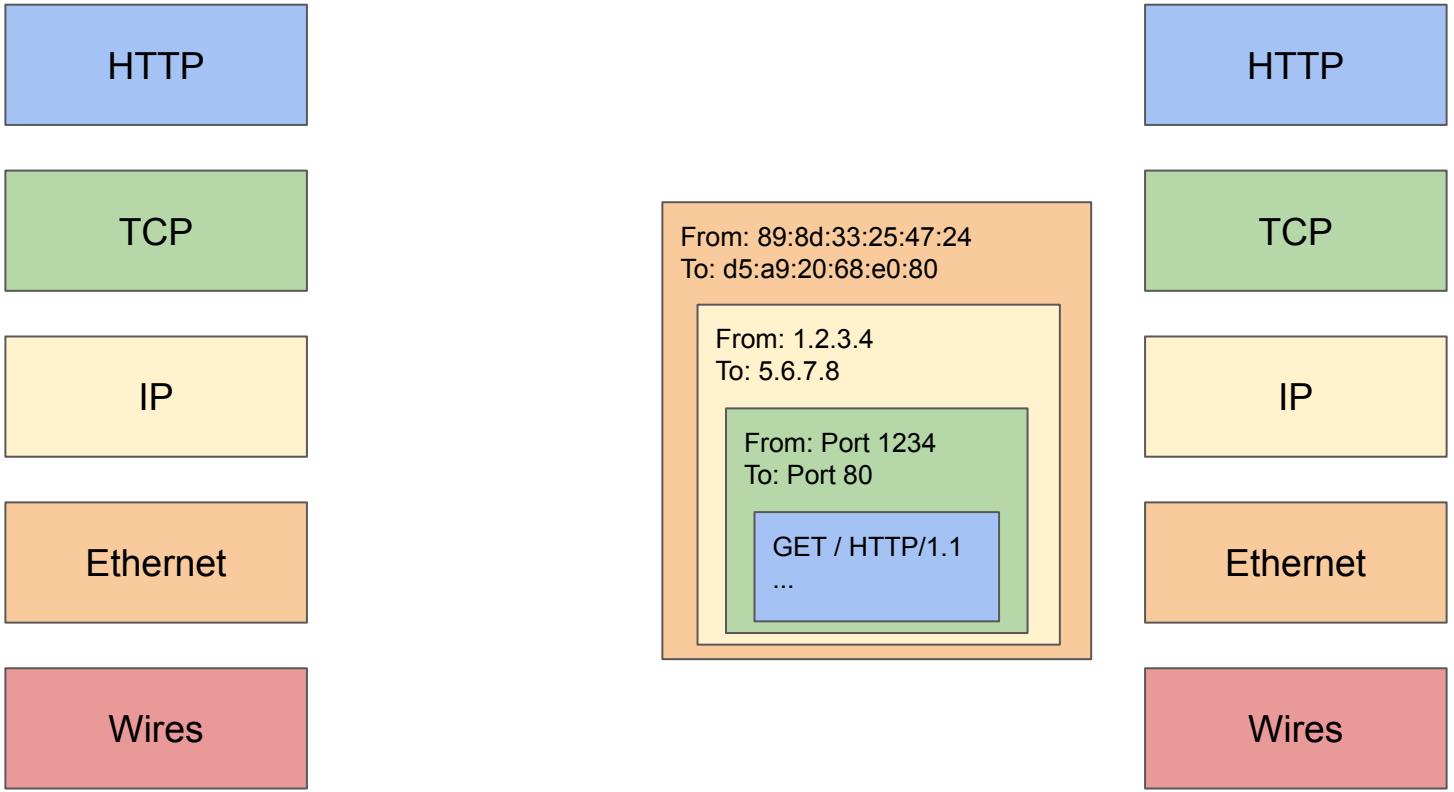
Example: HTTP Request



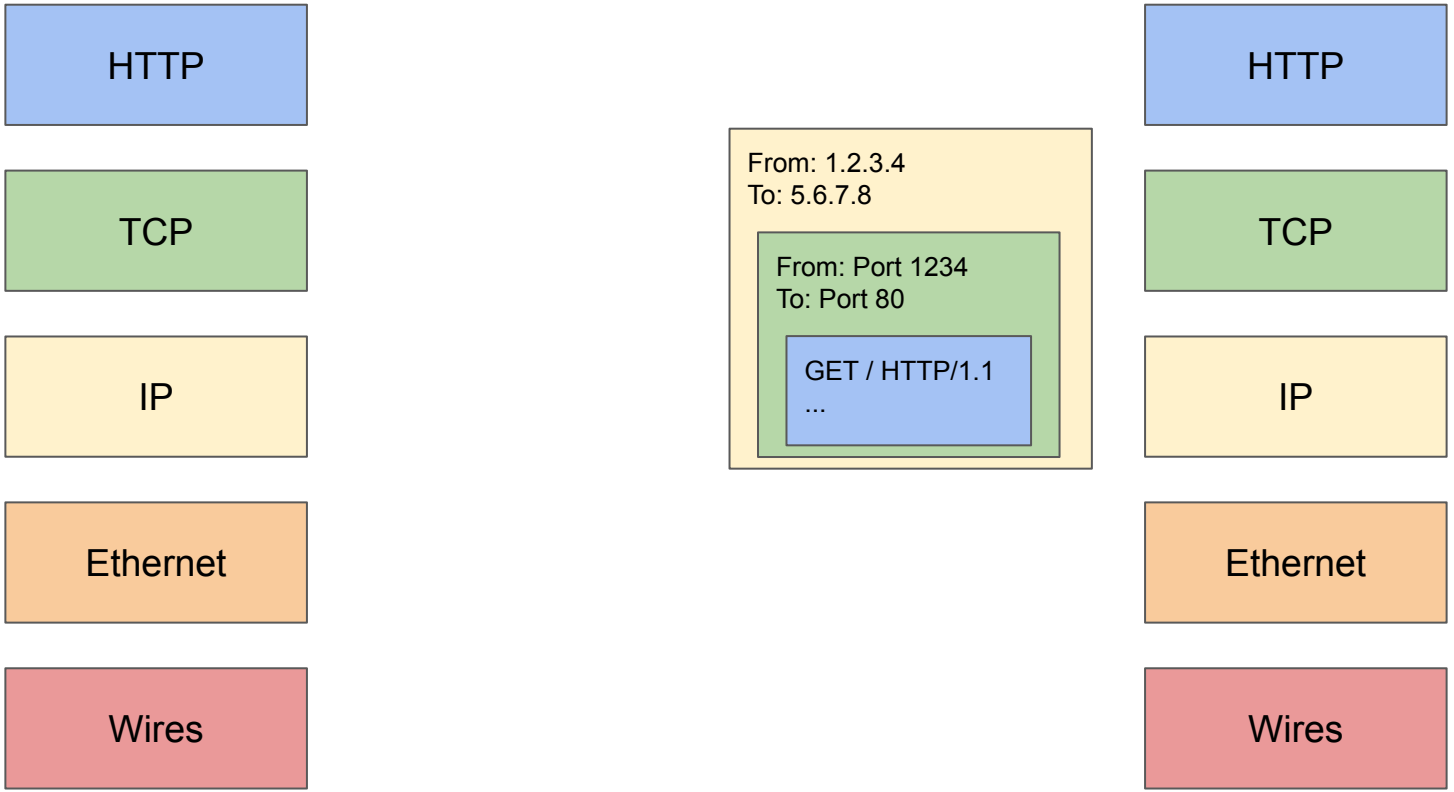
Example: HTTP Request



Example: HTTP Request



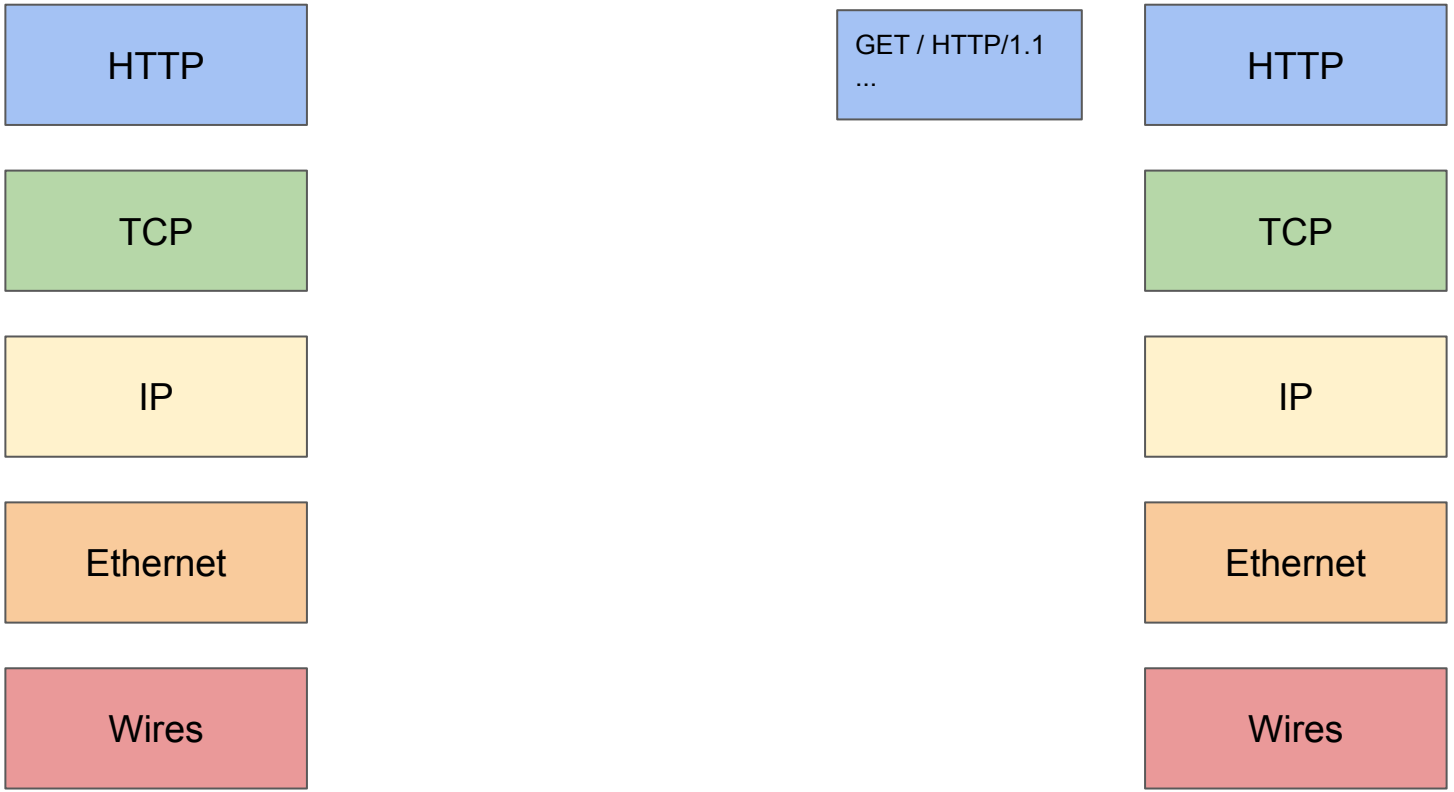
Example: HTTP Request



Example: HTTP Request



Example: HTTP Request



Show `wireshark`

Intro to Networking: Summary

- Internet: A global network of computers
  - Protocols: Agreed-upon systems of communication

- OSI model: A layered model of protocols
  - Layer 7: Applications and services (the web)
  - Layer 4: Transport of data (more on this next time)
  - Layer 3: Global packet delivery
    - IP: The universal Layer 3 protocol
    - IP addresses: 4-byte (or 16-byte) addressing system used by IP
  - Layer 2: Local frame delivery
    - Ethernet: The most common Layer 2 protocol
    - MAC addresses: 6-byte addressing system used by Ethernet
  - Layer 1: Physical communication of bits

